Table of Contents

Predicting Menstrual Hygiene Practice in India: Socioeconomic Inequality, Climate Exposure, and Spatial Generalization

Siddharth Tomar · Independent Researcher, Bengaluru, India · tomarsidharth62@gmail.com

Abstract

Background. Access to hygienic menstrual materials is a standard indicator of women's health in India, and ambient heat has been proposed as a driver of women's reproductive health outcomes. Existing NFHS-based quantitative work has focused on birth outcomes rather than menstrual practice, and the relative contribution of socioeconomic and climatic factors has not been quantified.

Methods. We linked individual records from India's National Family Health Survey (NFHS-5, 2019–21; 240,230 women aged 15–24) to two representations of climate: long-run normals from the DHS geospatial covariates, and an interview-windowed acute-heat exposure computed from ERA5-Land reanalysis. We trained logistic-regression and gradient-boosting models to predict exclusive use of hygienic materials, evaluated under both random and district-grouped ("spatial") cross-validation, and decomposed feature contributions by ablation. All outcome construction was validated against published NFHS-5 statistics.

Results. The gradient-boosting model reached a spatially-validated AUC of 0.758. Household wealth and the woman's education dominated prediction; long-run climate normals added +0.033 AUC and interview-windowed acute heat added +0.001. Random cross-validation overstated the flexible model's accuracy by 0.018 AUC through district memorization, an inflation absent in the linear model.

Conclusions. Menstrual hygiene practice in India is predicted overwhelmingly by socioeconomic status, not climate. We further demonstrate that standard cross-validation overstates model performance on spatially clustered health-survey data, and release an open, validated pipeline and district-level need map.

This report summarises the pipeline in this repository: linking NFHS-5 (2019–21) individual-level survey data to DHS geospatial climate covariates and Earth Engine–derived acute heat exposure, then modelling menstrual hygiene practice with an explicit random-vs-spatial cross-validation comparison.

This is predictive modelling, not causal inference. Nothing here estimates the *effect* of heat on hygiene practice. The question answered is narrower and more useful as an engineering artifact:

how much of a model’s apparent skill is genuine signal versus memorised geography, and which feature blocks carry that signal.

---

1. Data and pipeline

Stage	Source	Result
Survey recode	NFHS-5 individual recode (`IAIR7EFL.DTA`), women 15–24	241,180 respondents
Outcome construction	Multiple-response menstrual-material items, mapped by canonical label (not letter suffix — see `README.md`)	Four outcome definitions, 41%–78% prevalence spread
Climate normals	DHS Geospatial Covariates (`IAGC7AFL.csv`), cluster-level, 2000–2020 snapshots	99.1% of women matched to a covariate cluster
Acute heat exposure	ERA5-Land daily reanalysis via Google Earth Engine; district-relative 90th-percentile heat-index threshold; heat-day count in the 12 months before each woman’s interview	27,308 of 29,899 clusters computed (5 of 60 batches timed out and were skipped); heat exposure attached to 90.9% of women
Modelling frame	Survey + climate-normal + acute-heat features merged on cluster/district id	n = 240,230, 707 districts, 36 features

2. Recode validation

Before any modelling, the pipeline reproduces four externally published NFHS-5 figures directly from the raw recode. All four pass:

Check	Observed	Expected	Source
n, women 15–24	241,180	241,180	Singh & Singh 2025, *Front Reprod Health*
Any hygienic material (broad)	77.6%	77.6%	Singh & Singh 2025, Table 3
Exclusive hygienic material (broad)	50.0%	49.8%	Meher & Sahoo 2023, *BMC Women’s Health*
Sanitary napkin use	64.1%	64.4%	Singh & Singh 2025, Table 3

A recode that cannot reproduce known published numbers is not a recode worth modelling with. This one does.

Outcome definition is a modelling decision, not a detail

“Menstrual hygiene” has no single agreed definition. Two independent choices — whether locally prepared napkins count as hygienic, and whether *exclusive* use is required — produce four prevalence estimates spanning **41.0% to 77.6%**, a 28-point range wider than any effect this study (or comparable studies) could plausibly detect.

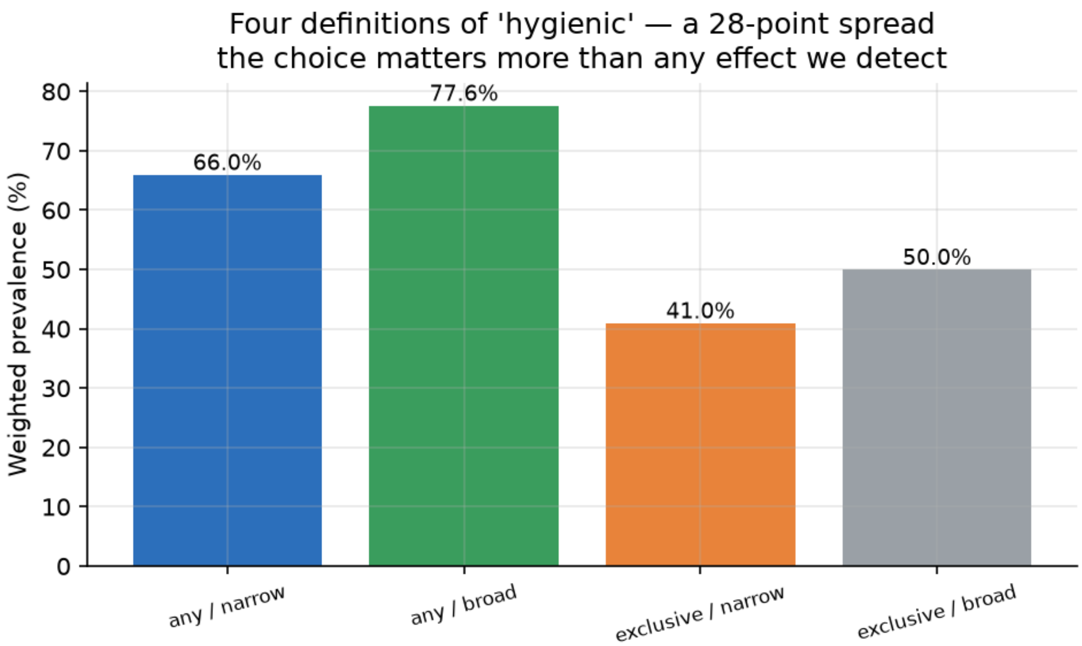

Figure 1

The modelling below uses **exclusive / narrow** (pads, tampons, or menstrual cup only — cloth and locally prepared napkins both disqualify) as the primary outcome, since it is the strictest and most defensible definition.

---

3. The real driver: wealth

Before climate enters the picture, hygienic-material use rises steeply and near-monotonically with household wealth quintile:

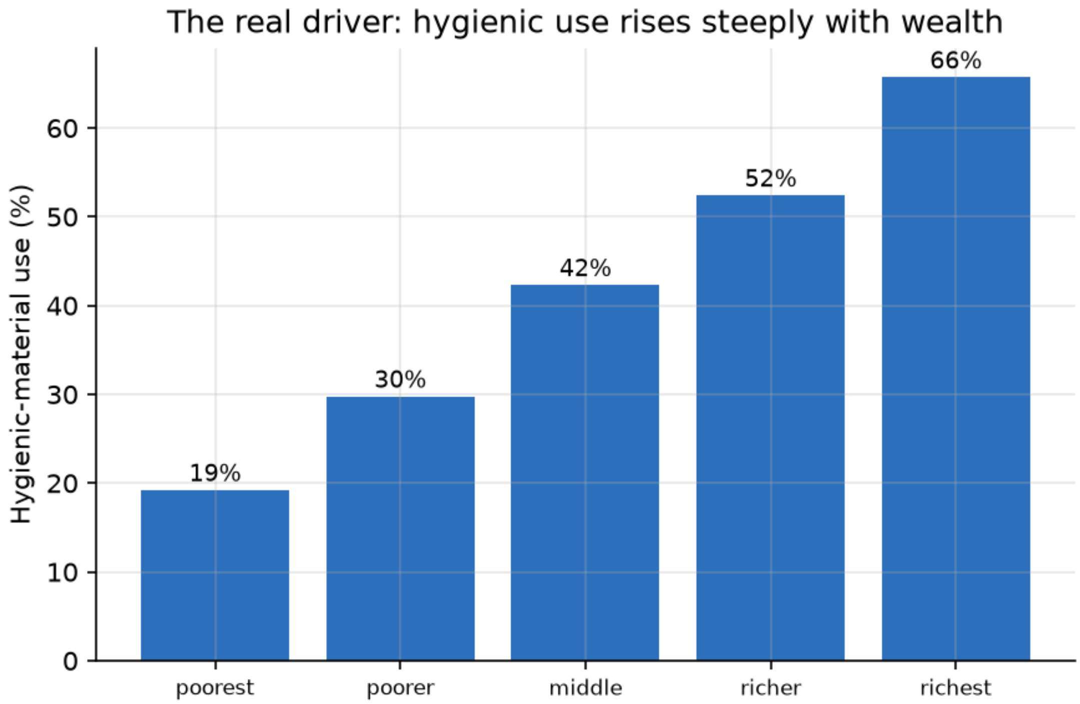

Figure 2

Any climate association has to be read against this gradient — wealth and climate normals are correlated across India's geography, which is exactly why the ablation in Section 5 separates them.

---

4. Random vs. spatial cross-validation

The central methodological claim of this repository: **random k-fold CV overstates predictive skill on spatially clustered data**, because test-set women live in the same districts as training-set women, so the model partly gets credit for memorising district-level baselines rather than learning transferable signal. District-grouped ("spatial") CV — where every test district is entirely unseen during training — removes that credit.

Model	Random-CV AUC	Spatial-CV AUC	Leakage gap
Logistic regression	0.729	0.727	0.002
Gradient boosting	0.776	0.758	0.018

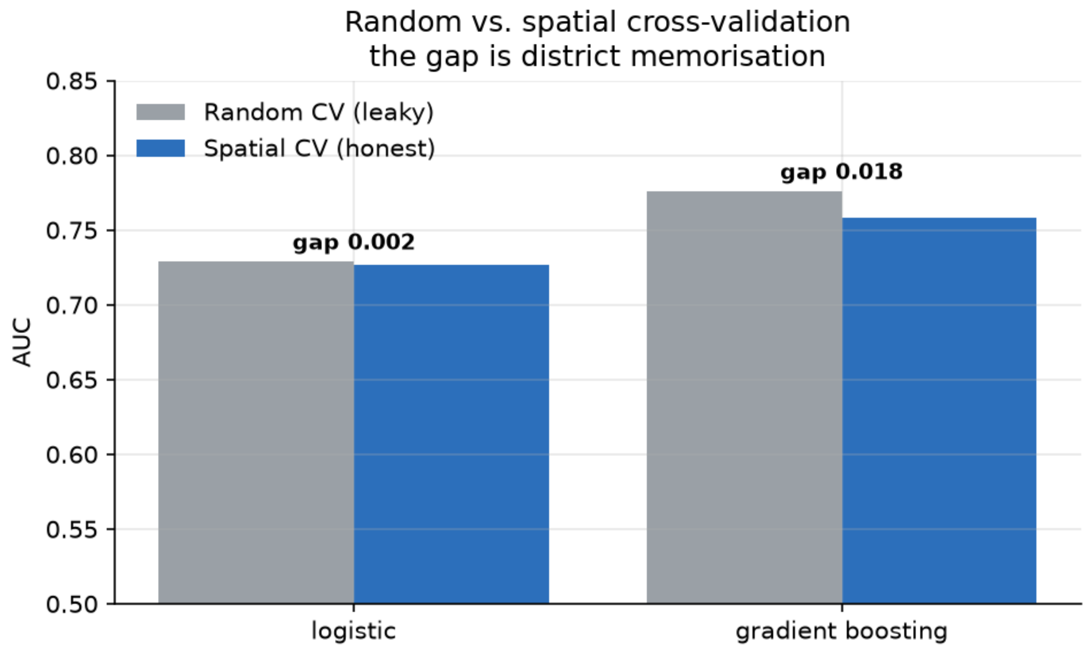

Figure 3

The logistic model barely leaks (0.002) — as a linear model with modest capacity, it has little room to memorise district idiosyncrasies. Gradient boosting leaks nearly an order of magnitude more (0.018): its extra flexibility is partly spent fitting district-specific patterns that don't generalise to new districts. **The honest number for gradient boosting is spatial AUC 0.758, not the random-CV 0.776** a less careful evaluation would report.

Calibration

Out-of-fold, spatial-CV predicted probabilities track observed rates reasonably closely across the probability range:

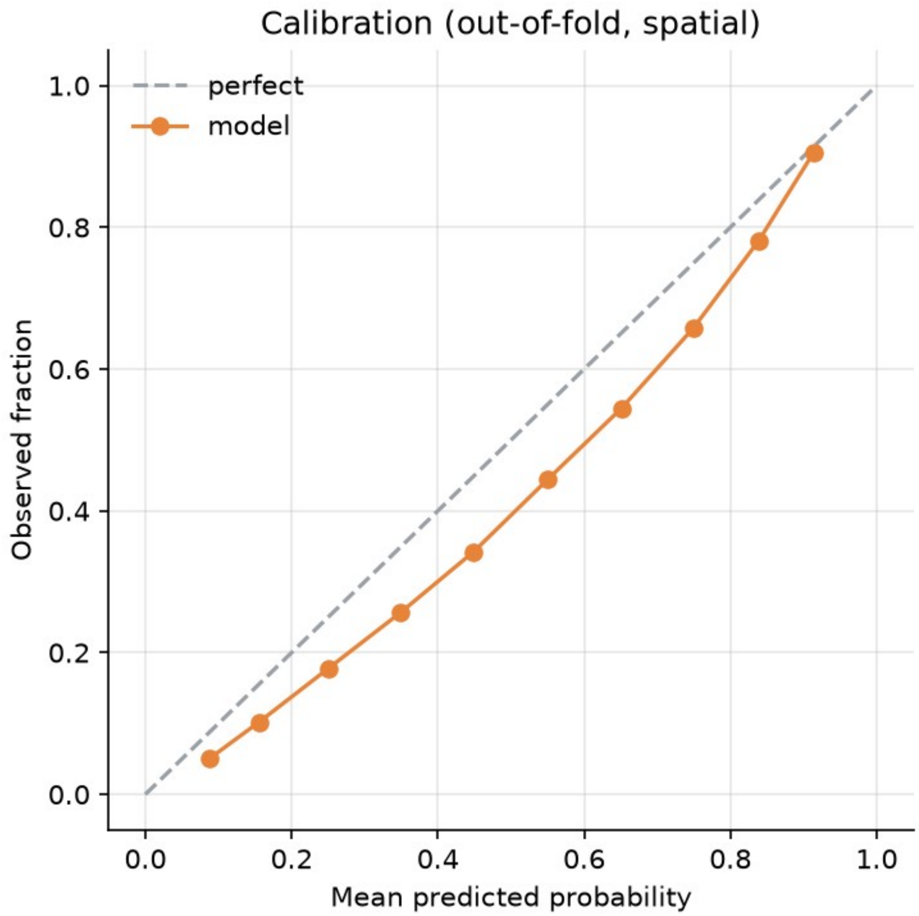

Figure 4

5. Feature ablation: what does climate actually add?

Holding the model (gradient boosting, spatial CV) fixed, three feature blocks are compared:

Block	Spatial AUC	Δ vs. previous
SES only	0.724	—
+ DHS climate normals (typical district climate)	0.757	**+3.3 pts**
+ acute heat (ERA5-Land, interview-windowed heat days)	0.758	+0.1 pts

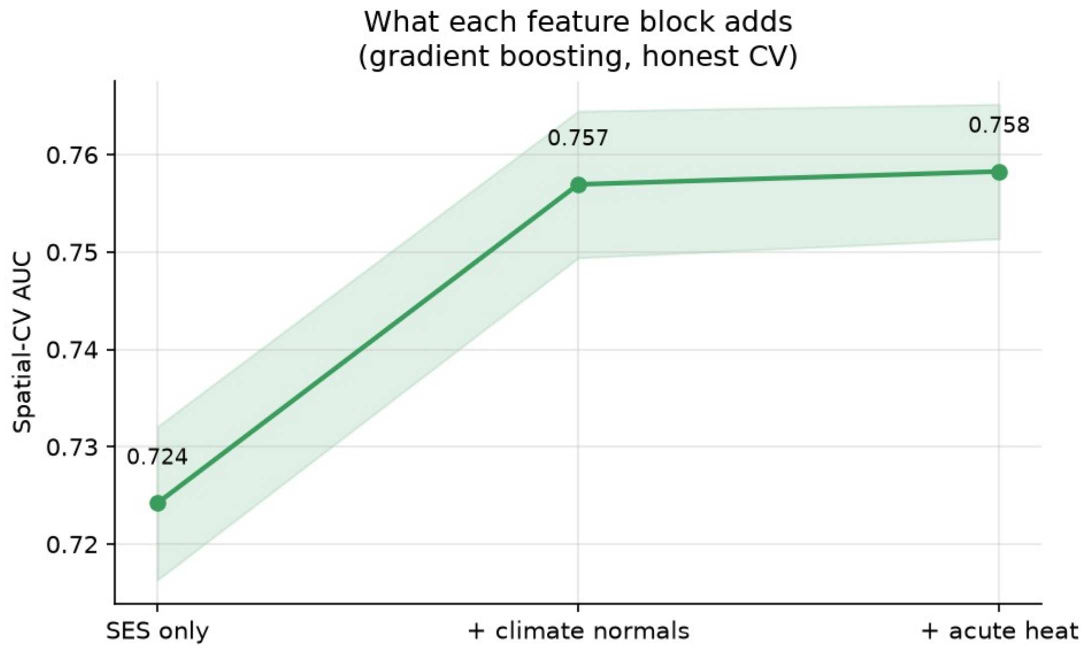

Figure 5

This is the most notable empirical finding in the pipeline. Climate **normals** — a district's typical thermal environment, available with no Earth Engine dependency — carry almost all the climate-related predictive signal. The **acute**, interview-windowed heat-day count, despite requiring the most engineering effort in this project (dewpoint band-name fixes, batching, null-threshold handling, reducer output naming — see commit history), adds essentially nothing on top. Two readings are possible and are not distinguished by this design: acute heat genuinely carries little independent signal for this outcome, or district-normal climate is already a good enough proxy for a district's acute heat regime that the acute measure is largely redundant with it.

What drives the prediction

SHAP importance for the gradient-boosting model, full feature set:

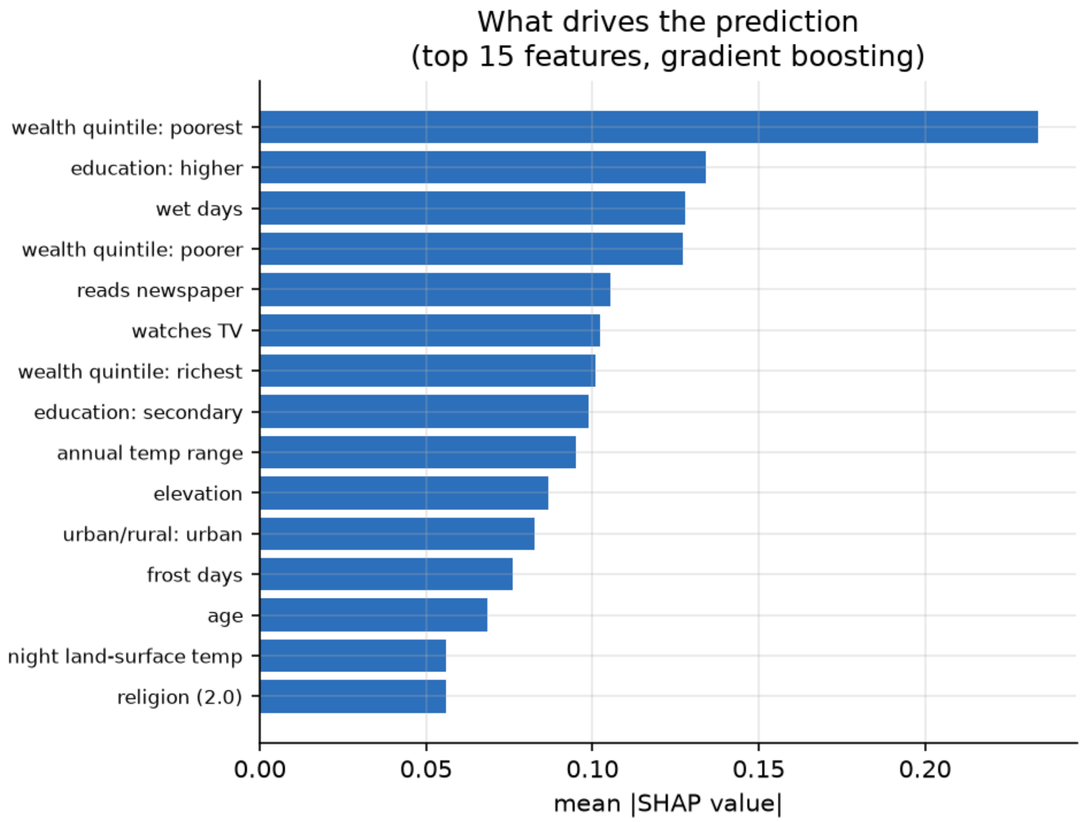

Figure 6

---

6. District need map

Districts ranked by weighted hygienic-material-use rate (raw need), with an adjusted-need flag: **negative adjusted gap = the district does worse than its own socioeconomic profile predicts** (out-of-fold, spatially cross-validated prediction, so this is never in-sample). Full table: `outputs/tables/district_need_map.csv` (707 districts, all with 270+ respondents — district-level aggregates only, no individual data).

Top 15 highest-need districts:

rank	district	state	n women	hygienic rate	adjusted gap
1	South Salmara Mancachar	Assam	382	6.7%	−0.164
2	Sidhi	Madhya Pradesh	453	6.9%	−0.209
3	Karimganj	Assam	415	7.6%	−0.173
4	Purba	Bihar	473	7.8%	−0.141

rank	district	state	n women	hygienic rate	adjusted gap
	Champaran				
5	Mahisagar	Gujarat	294	8.4%	−0.217
6	Banda	Uttar Pradesh	435	8.4%	−0.163
7	Dohad	Gujarat	443	9.1%	−0.183
8	Cachar	Assam	379	9.8%	−0.204
9	Gopalganj	Bihar	494	10.3%	−0.157
10	East Jaintia Hills	Meghalaya	500	10.4%	−0.297
11	Hardoi	Uttar Pradesh	511	10.6%	−0.099
12	Siwan	Bihar	528	10.6%	−0.136
13	Chirang	Assam	370	10.6%	−0.177
14	West Jaintia Hills	Meghalaya	420	10.9%	−0.250
15	Mon	Nagaland	273	11.0%	−0.346

Every one of the top-15 districts is flagged underperforming. Assam accounts for 4 of the 15; the two Jaintia Hills districts (Meghalaya) and Mon (Nagaland) show the largest adjusted-need gaps, meaning something beyond their SES profile — not captured by wealth, education, or water/sanitation access alone — is holding them back further than poverty would predict on its own.

This need map is presented as a ranked table rather than a choropleth by deliberate choice: openly available district-boundary shapefiles are built on 2011 Census boundaries (641 districts), while NFHS-5 uses the 707 post-2011 districts, so roughly 127 newer districts — including several of the highest-need ones here, such as South Salmara Mancachar and both Jaintia Hills districts — have no corresponding 2011 polygon; mapping would require merging them back into their pre-split parent districts, losing exactly the district-level resolution this ranking depends on.

---

7. Limitations

1. **Predictive, not causal.** No claim is made that heat, wealth, or any other feature *causes* hygiene practice. AUC and SHAP importance describe association and predictive contribution, not effect size.
2. **Cross-sectional design.** NFHS-5 is a single interview wave; temporality between any exposure and the outcome cannot be established.

3. **Ecological climate features.** DHS geospatial covariates and the acute heat-day count are cluster/location-level, not measures of an individual woman's personal exposure (occupational, indoor, commuting exposure are unmeasured).
4. **Outcome definition sensitivity.** The primary outcome (exclusive/narrow) is one of four defensible definitions spanning a 28-point prevalence range (Section 2); results should be read as conditional on this choice.
5. **Acute heat coverage gap.** 5 of 60 Earth Engine batches (≈2,591 of 29,899 clusters) timed out and were skipped rather than retried, giving 90.9% rather than 100% heat-exposure coverage. The skipped clusters are disproportionately coastal or remote, so this missingness is spatially structured rather than random; missing acute-heat values were median-imputed within cross-validation folds. Because the acute-heat block contributes negligibly to predictive performance (+0.1 pts AUC), the imputation choice has no material effect on the results.
6. **Self-reported outcome.** Menstrual material use is self-reported in interview, with the usual caveats about social-desirability and recall.

---

8. Reproducing this report

```
pip install -r requirements.txt
python scripts/01_build_survey.py --recode data/raw/IAIR7EFL.DTA
python scripts/02a_merge_covariates.py
earthengine authenticate
python scripts/02b_gee_heat.py --gps data/raw/IAGE7AFL.shp
python scripts/03_train.py
python scripts/04_district_map.py
python scripts/05_figures.py
```

NFHS-5 microdata is not redistributed in this repository (DHS terms); obtain it from The DHS Program. All figures and result tables in `outputs/` are generated artifacts and are tracked in this repo; raw and intermediate data are not.

Declarations

Data availability. All code, figures, and summary tables are at github.com/sidharthtomar/nfhs-climate-link. NFHS microdata are not redistributed per DHS terms and are freely available on registration from The DHS Program (dhsprogram.com). ERA5-Land data are from the Copernicus Climate Data Store.

Ethics. This study is a secondary analysis of publicly available, de-identified survey data. The original NFHS-5 protocol received ethical approval from the IIPS and ICF Institutional Review Boards, and informed consent was obtained from all participants at the time of the survey. No further ethical approval was required for this secondary analysis; no attempt was made to re-identify individuals or communities.

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Contact. tomarsidharth62@gmail.com